

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & M_{xxz} \\ 0 & -M_{xxy} & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ M_{xxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = \mathbb{M}_{3b}^{(1)}$$

$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)}$$

$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)}$$

$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ -M_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzy} = \mathbb{M}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxxx} \\ M_{xxxx} & -M_{xxxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxxx} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4b}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzx} & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & M_{xxxy} & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = -2\mathbb{T}_{3b}^{(1)}$$

$$M_{xxxy} = 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & M_{xxxy} \\ 0 & -M_{xxzx} & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{T}_{3b}^{(2)}$$

$$M_{xxxy} = \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzx} = \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{yzxy} = \mathbb{T}_{3b}^{(3)}$$

$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} & 0 & M_{xxxz} & M_{xxyx} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxxy} & 0 & -M_{xxxz} & -M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} & 0 & M_{yzzx} & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & -M_{yzzx} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -\frac{M_{xxxxy}}{2} - \frac{M_{xxxxy}}{2} & \frac{M_{xxxxy}}{2} + \frac{M_{xxxxy}}{2} & 0 & -M_{xxxz} & 0 & 0 & -M_{xxxz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} + \mathbb{T}_{3b}^{(2)} \\
M_{xxxy} &= 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{xxxz} &= \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} - \mathbb{T}_{3b}^{(2)} \\
M_{xyyx} &= 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \\
M_{zzzy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yzzx} &= -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{yzxy} &= -\mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} \\
M_{yxxz} &= -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} & 0 & M_{yxxz} & M_{yzxy} \\
M_{yzxy} & -M_{yzxy} & 0 & M_{yxxz} & 0 & 0 & M_{yzzx} & 0 & 0 \\
M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{yzzx} &= -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{yzxy} &= \mathbb{T}_{3b}^{(3)} \\
M_{yxxz} &= \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyzz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & M_{xxxx} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
-M_{xxxx} & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
M_{xxxx} & 0 & 0 \\
0 & 0 & -M_{xxxx} \\
0 & -M_{xxxx} & 0
\end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4b}^{(1)}$$